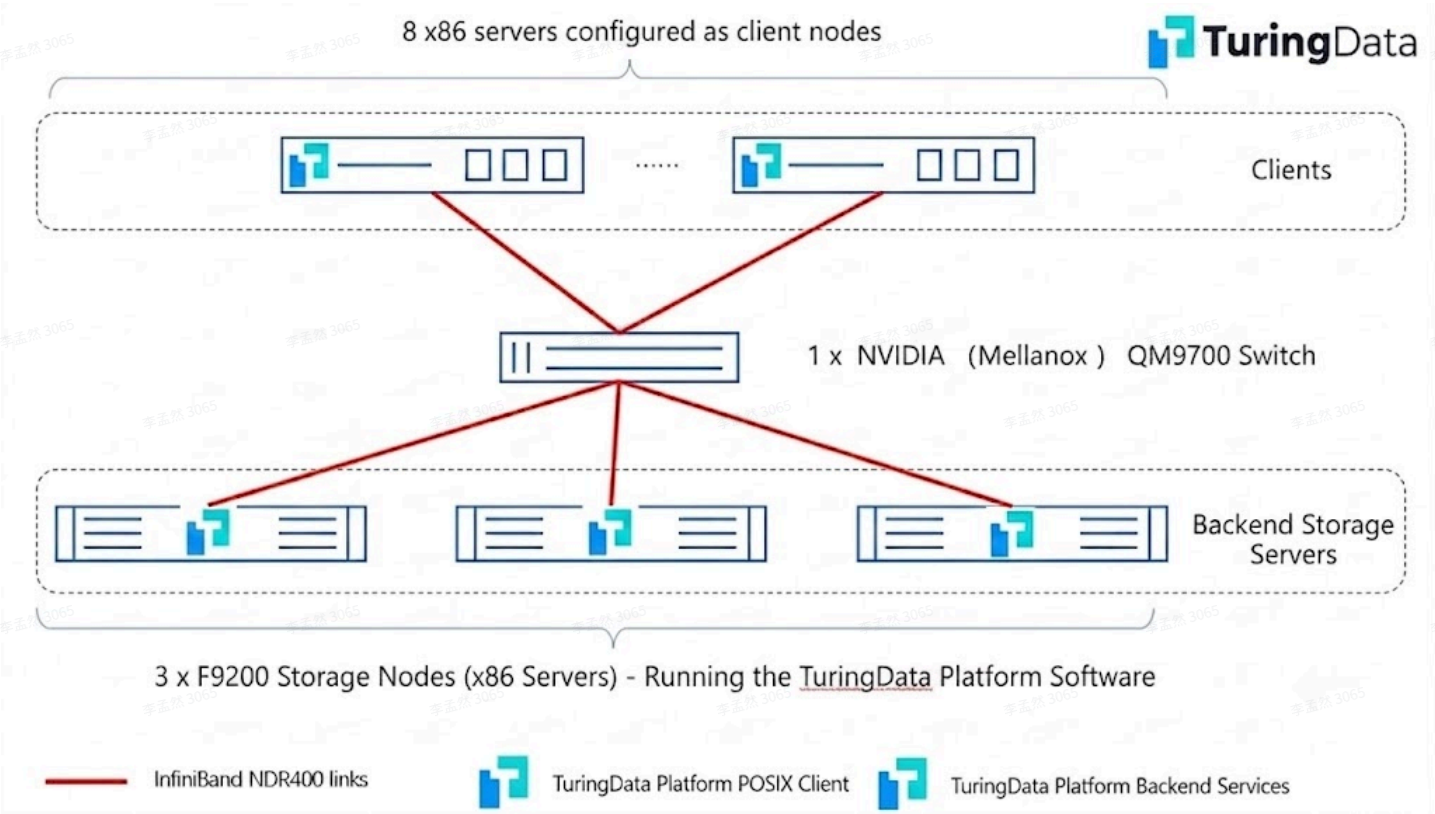


TuringData_F9200_8Clients

1. Hardware

The hardware configuration includes eight clients and three servers interconnected via a shared switch.

Hardware Diagram



1.1 Client

The client is a commercial x86-architecture server node featuring a dual-socket (2P) Intel-based motherboard, 512GB of RAM, and default out-of-the-box settings. It communicates with the server over InfiniBand, utilizing two dedicated links exclusively for storage traffic.

The detailed configuration of the client is shown in the following table:

Client Node components	Configuration Description	Client Node Count
Client CPU	INTEL(R) XEON(R) PLATINUM 6530 x 2	8
Client Memory	512GB DDR5 (16 x 32GB RDIMMs)	

Client OS Boot Drive	480GB SSD x 1	
Client network_type	Mellanox CX7 Single-port NDR400 InfiniBand HCAs x 2	

1.2 Server

The test environment utilizes a TuringData F9200 storage appliance, an x86-based commercial storage solution running the TuringData distributed file system. The system comprises three storage nodes, each configured with: one 480GB SATA SSD for the operating system, sixteen Solidigm 3.84 TB NVMe SSDs for data and metadata storage, and four single-port ConnectX-7 400Gbps NDR InfiniBand host channel adapters for high-speed connectivity.

The server communicate with the client through IB:

- 4x single-port NDR400 adapters for a total of 4x NDR400 links

The detailed configuration of the servers is shown in the following table:

Server Node components	Configuration Description	Server Node Count
Server CPU	Intel® Xeon® 6740E Processor x 2	3
Server Memory	512GB DDR5	
Server OS Boot Drive	480GB SATA SSD x 1	
data and metadata drives	Solidigm 3.84TB nvme SSD x 16	
Server network_type	Mellanox CX7 Single-port NDR400 InfiniBand HCAs x 4	

1.3 Switch

The switch is a commercial NVIDIA （Mellanox） QM9700 The client and server is connected to it.

2. Software

2.1 Client

The configuration of the client-side software is shown in the table below:

Software Type	Software Version

Client OS	Ubuntu 24.04 LTS
Client Kernel	6.8.0-60-generic
client	TuringData Platform 3.1.0
Mellanox OFED version	MLNX_OFED_LINUX-24.10-4.1.4.0

2.2 Server

The configuration of the Server-side software is shown in the table below:

Software Type	Software Version
TuringData Platform OS	TuringData Platform 3.1.0
Storage Servers OS	Rocky Linux release 9.6 (Blue Onyx)
Storage Servers Kernel	5.14.0-570.17.1.el9_6.x86_64
Mellanox OFED version	MLNX_OFED_LINUX-24.10-4.1.4.0

3. Settings

3.1 Client

The standard client (Version: TuringData Platform 3.1.0) is installed on the client using the default installation

procedure. client is used for the client to access the remote TuringData Platform filesystem.

3.2 Server

For detailed configuration information about storage servers, see the table below:

storage_system	Description
vendor_name	TuringData
solution_name	TuringData Platform
model_name	F9200

version	TuringData Platform 3.1.0			
usable_capacity_TB	80			
raw_capacityy_TB	160			

3.3 Switch

No tuning applied.